Sound speed and resonance

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel time and resonant conditions to interpret sound.

SUCCESS CRITERIA - I can...

1. Account for round-trip distance in an echo.
2. Identify resonance as a response near a natural frequency.
3. Use an appropriate wavelength relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

At fixed wave speed, frequency becomes 3 times as large. Compare wavelength and typical diffraction through the same opening.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $d_{echo} = v\,\Delta t/2$; $v = 340$ m/s.
Echo time is a round trip. Resonance occurs near a natural mode under suitable damping.

Worked reasoning: In an ideal tube open at both ends, the fundamental wavelength is twice its length. For L=0.5 m and sound speed 340 m/s, f=340/1=340 Hz.

Your turn: Sketch the fundamental with displacement antinodes at both ends.

open-open tube

_____

_____

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: ① ② ③

03 Read and interpret the model

An echo returns from a wall. Draw the outgoing and returning path. Why does sound's total travel distance exceed wall range?



Marzano check - circle: ① ② ③

04 Apply the physics

In air with sound speed 340 m/s, an echo returns after 0.30 s. Find the wall distance. In a separate instrument model, an open-open pipe is 0.50 m long. Use $\lambda = 2L$ and $f = v/\lambda$ to find its fundamental wavelength and frequency.

Measurement	Path / wavelength	Result
Echo: 0.30 s	Out and back	____ m
Open pipe: 0.50 m	$\lambda = 2L$	____ Hz

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why does a resonant response not mean that any object can respond strongly to every frequency?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A heat engine absorbs Q_{hot}=3500 J and rejects Q_{cold}=2450 J each cycle. Find work and efficiency. Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

A closed-open pipe fundamental has L=0.25 m. Use λ=4 L and v=340 to find frequency.

closed-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.85 m tube is closed at one end and open at the other. Use v=340 m/s and λ=4 L for its fundamental. Find the first three allowed ideal resonance frequencies.

closed-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3